Team contract

DIT356 H24 Mini Project: Distributed Systems Development, Team 11

Project Goals

1. **Grade Objective**: Our target grade for this project is a 5. We will focus on achieving excellence across four key dimensions: Technical Quality, Architecture, Development Process, and Individual Contribution.
2. **Individual Goals**:
    - **Loura**: Focus on working with databases and MQTT.
    - **Nika**: Gain experience in backend and database work.
    - **Rene**: Open to any area, willing to adapt as needed.
    - **Georgi**: Emphasis on management and architectural aspects.
    - **Reihaneh**: Seeks involvement across all areas.
3. **Skills Development**: We aim to challenge ourselves, enhance technical and management skills, and gain hands-on experience with tools, methodologies, and technologies specified in our project.

Project Planning

1. **Timeline and Milestones**: We will follow pre-defined milestones, ensuring tasks are broken down into smaller goals with deadlines that allow a safety margin before final submissions.
2. **Meetings**:
    - **Weekly TA Meetings**: Wednesdays or Thursdays.
    - **Team Meetings**: Two to three per week, primarily in-person on campus, with emergency meetings held online if needed.
3. **Task Assignment**: Tasks will be assigned and deadlines set collaboratively to balance workloads and ensure progress toward milestones.

Communication and Collaboration

1. **Response Time**:
    - **Reyhaneh**: Response within 2 hours during the week; limited availability on weekends(unless close to deadline).
    - **Nika, Loura, Rene, and Georgi**: Generally available anytime.
2. **Tools**:
    - **Communication**: Discord for day-to-day messaging and coordination.
    - **Task Management**: GitLab for code management, issue tracking, and code reviews; Miro for project planning and management visualization.

Technologies

Core Technologies:

- **Middleware:** Use MQTT/messaging for efficient communication across system components.
- **Web Interface:** Vue3.js will be used to build a responsive, user-friendly, web-based interface.
- **Database:** MongoDB will serve as our main database, supporting flexible and scalable data storage.
- **Networking Engine:** Netty will be implemented as our networking engine to handle high-performance, asynchronous communication.
- **Programming Language:** Java will be used as the primary language for developing middleware components.
- **Message Broker:** Eclipse Mosquitto will serve as the MQTT message broker, facilitating pub-sub messaging.
- **MQTT Client Library:** Eclipse Paho will be utilized as the MQTT client library, enabling communication with the message broker.
- **Responsiveness:** Bootstrap will ensure that our web-based interface is responsive and visually accessible on multiple devices.

Development Process

1. **Agile Methodology**:
   - **Sprint Duration**: Defined based on assignment schedules.
   - **Sprint Planning**: Meetings will occur on the assignment release day.
   - **Sprint Retrospectives**: Held on either Friday or Monday.
   - **Stand-Up Frequency**: Three to five stand-ups weekly, in person or on Discord.
2. **Task Assignment**: Tasks will be assigned within each sprint, with roles rotating for sprint management.
3. **Quality Assurance**: All code will undergo peer review, with a dedicated person reviewing branches they did not contribute to.
4. **Code Contributions**:
   - **Branching Strategy**: A developer branch will serve as the main duplicate. Tasks will be broken down into branches associated with individual issues.
   - **Commit Standards**: A standardized commit template will be followed.
   - **Pair Programming**: To be used as necessary for complex tasks or to resolve challenges.

Roles and Responsibilities

1. **Sprint Manager**: Each sprint will have a designated manager to monitor task progress and ensure team members are supported. Georgi will manage Sprint 1.

2. **General Responsibilities**: All team members will contribute actively to task discussions, assist others when needed, and participate in code review and quality assurance processes.

Handling Deviations

1. **Process Deviations**: In case of any delays or blocking tasks, deadlines will be set with a buffer to manage unexpected issues. Urgent or critical issues will be addressed in additional meetings if needed.
2. **Problem Resolution**: If a member falls behind or encounters challenges, the team will work collaboratively to redistribute tasks or adjust timelines.

Architecture and Documentation

1. **Architectural Patterns**:

- **Microservices**: Our system will be designed as a collection of independent services, each responsible for a specific function, allowing for scalability, modularity, and easier maintenance.
- **Publish-Subscribe (Pub-Sub)**: We will implement a pub-sub pattern using MQTT for efficient, asynchronous message passing between components. This will support real-time communication and decoupling of message producers and consumers.
- **Client-Server**: The system will follow a client-server architecture to manage communication between user interfaces (clients) and backend services (servers), ensuring a clear separation of responsibilities and improving system manageability.

2. **Documentation Standards**:

- **4+1 View Model**: We will use the 4+1 View Model to structure our architectural documentation, covering Logical, Development, Process, Physical, and Scenario views.

View Name	Corresponding UML Diagrams
Logical View	Class Diagram(Rene)
Process View	Sequence Diagram(Georgi)
Physical View	Deployment Diagram(Laura)
Development View	Component Diagram(Reihaneh)

Use Case View	Use Case Diagram(Nika)

- **Clear Documentation**: All architectural decisions and system designs will be documented to maintain clarity and ensure easy understanding for all team members and stakeholders.

Contingency Plans

1. **Back-up Assignments**: In case of unforeseen issues with task completion, tasks may be reassigned or deadlines adjusted to ensure project continuity.
2. **Communication**: We commit to proactive communication to address concerns early, minimizing the impact on our timeline and quality.

Final Notes

- We will continuously evaluate and refine our processes to enhance efficiency and maintain a collaborative, supportive environment.
- Documentation will be maintained for all decisions, processes, and deviations, fostering transparency and accountability.

Signatures:

Loura:



Nika:



Rene:



Georgi:



Reyhaneh:

